# SECTION 18 STEELWORK

## GENERAL

***Requirements*** 18.01

(8) All measurements shall be made by means of an approved steel tape or other instrument. Dimensions of all steelwork given on the Drawings are assumed to be taken at a temperature of 20℃.

(9) All labour, equipment and materials necessary to perform surface preparation and coating shall be provided in accordance with these specifications. All personnel involved in surface preparation or coating work shall be thoroughly trained as to the materials and specifications pertaining to the coating systems.

(10) The coating system specified herein shall fully comply with any local and/or regional air quality or Volatile Organic Content (VOC) emission standards or requirements. Should modifications or alternate coating systems be required to meet such standards or regulations, full details and documentation shall be provided of the proposed alternate system for acceptance.

(11) The coating system specified herein shall fully comply with any local and/or regional air quality or Volatile Organic Content (VOC) emission standards or requirements. Should modifications or alternate coating systems be required to meet such standards or regulations, full details and documentation shall be provided of the proposed alternate system for acceptance.

(12) Product data sheets and specifications for mixing, application and curing shall be considered an integral part of this specification. Where a conflict arises between the requirements of this specification and a manufacturer's instructions the more onerous conditions shall apply.

(13) The coating systems to be used are specified in GS Appendix 18.2. The coating systems shall have a minimum service life of 30 years and the *Contractor* shall specify the intervals between the required maintenance. The *Contractor* shall also provide a 15 years warranty for the paint systems.

(14) If a coating option is indicated, the selected option shall be used as specified throughout the coating operation in accordance with these Specifications.

(15) A technical representative from the coating manufacturer shall be present to inspect and witness the application of each coating system throughout the duration of the work.

(16) Succeeding coats of paint shall be of a colour that contrasts with the colour of the previous coat.

(17) Pre-painted equipment incorporated into the work shall be recoated if the paint system does not meet or exceed the

requirements of this specification.

(18) The following surfaces shall not be grit blasted, coated or painted unless otherwise specified: aluminium, brass, rubber, glass, plastic, concrete, stainless steel, fiberglass reinforced plastic, PVC, stainless steel weather jacketing, insulation, plated surfaces, machined flange faces or instrument cases.

(19) Not used.

(20) Not used.

(21) Protection of steelwork against corrosion shall comply with BS EN ISO 12944 and BS EN ISO 14713 or equivalent accepted by the *Supervisor*. Unless otherwise specified on the Drawings, all steelwork shall be hot-dip galvanized to BS EN ISO 1461 by Specialist Contractor List of accepted Suppliers of Materials and Specialist Contractors for Public Works and shall in addition be painted.

***Hot Dip Galvanizing*** 18.01A All steel components shall be hot dip galvanized in accordance with BS EN ISO 1461 : 2009 after fabrication by specialist contractors in the "Class V: Hot dip galvanizing" of the "Specialized Operations for Highway Structure" category of the List of Approved Suppliers of Materials and Specialist Contractors of Public Works. Steel hollow sections shall be sealed wherever this can be done without affecting the galvanizing process. If venting is necessary, the vents shall be carefully detailed and positioned so as to be inconspicuous or be effectively scaled immediately after galvanizing. The minimum thickness of hot-dip galvanizing coating shall be 85um,

**CONSITITUENT PRODUCTS**

***Structural steel*** 18.12A (1) Hot rolled sections complying with BS 4: Part 1: 2005, BS EN 10025, BS EN 10210-1&2 : 2006, BS EN 10034 : 1993, BS EN 10056-1: 1999, BS EN 10056-2 : 1993, BS EN 10067 : 1997 shall not be replaced with sections complying with other standards unless accepted by the *Supervisor*; if accepted, the sections shall have equivalent properties to, and the dimensional tolerances shall comply with, the relevant British Standard.

(2) All materials and components shall be ordered from and produced by manufacturers and suppliers with proven and documented experience of producing material to the codes and standards required in the following specification:

- Structural steel shall in general comply with BS 5400 : Part 6, Section 3.1
- Structural steel rolled shapes and plates shall be thermo-mechanical rolled steel according to BS EN 10025 Part 3 / Part 4 and :
- BS EN 10025 Part 4 for plate thickness less than or

equal to 40mm.

- BS EN 10025 Part 4 for plate thickness greater than 40mm.
- Structural steel for parapets, lamp posts, sign gantries, gantry rails and other components specified on the Drawings as S355 steel shall be normalized rolled steel according to BS EN 10025 Part 3 / Part 4 and:
- BS EN 10025 Part 3 for plate thickness less than or equal to 40mm.
- BS EN 10025 Part 3 for plate thickness greater than 40mm.

(3) The following additional requirements apply:

The applied steel grades and qualities shall be stated on the *Contractor*'s shop drawings, where grade and quality class shall be indicated. The shop weld joints and site weld joints shall be identified and indicated in the *Contractor*'s shop drawings.

Option 3) Product analysis shall be carried out on two samples for each cast. Chemical composition of the ladle analysis shall be according to values for BS EN 10025. Permissible deviations of the analysis shall be according to BS EN 10025, Table 1.

The test results shall be documented in the certificates of materials.

The steel requiring above documentation is identified on plans as "z-steel".

Option 10) applies with no hard steel stamping for steel delivered for incorporation in the works. Steel shall be identified by other suitable marking. The marking shall enable the material to be clearly identified and checked against the documentation which must include details of origin, methods of production, composition of material, inspection reports and test certificates. The marking shall be according to the rules in BS EN 10113 and BS EN 10204, so that a statement of this marking on the certificate is sufficient for an unambiguous identification of the steel.

(4) Steelwork delivered for incorporation in the works shall be identified by marking. No hard stamp is allowed for any of the steel delivered for incorporation in the works.

***Structural steel products - General*** 18.13***S*** (1) Flat rolled stainless steel shall comply with BS EN 10029, BS EN 10048, BS EN 10051 + A1, BS EN 10095 and BS EN ISO 9445, Grade 1.4436 in the softened condition.

(2) Wrought stainless steel shall comply with BS EN 10084, BS EN 10085, BS EN 10087, BS EN 10088, BS EN10095, BS EN 10250-4 and PD 970, Grade 1.4436.

***Welding consumables*** 18.16

GS Clause 18.16 is changed to GS Clause 18.16(1).

The following is added after GS Clause 18.16(1) :-

(2) Welding of steel shall be in accordance with BS EN 1011 -1 and BS EN 1011 -2 and electrodes shall be in accordance with BS EN ISO 2560:2009, BS EN ISO 14341:2008, BS EN ISO 14171:2010, BS EN ISO 14174:2012, BS EN ISO 14343: 2009. Welding consumables used in the fusion weld of steel casting shall comply with BS 4570.

***Mechanical fasteners – Structural bolting assemblies for nonpreloaded applications*** 18.18***S***

Stainless steel bolts and nuts shall comply with BS EN ISO 3506-1 & 2, steel Grade A4 and property class 80. Stainless steel washers shall comply with BS EN 10029, BS EN 10048, BS EN 10051:2010 + A1, BS EN 10095 and BS EN ISO 9445, Grade 1.4436 in the softened condition. The dimensions and tolerances of bolts and nuts shall comply with BS 3692. The dimensions and tolerances of tapered washers shall comply with BS 3410 and the dimensions and tolerances of flat washers shall comply with BS 4320, Form C.

**WELDING**

***Qualification of welding procedures and welding personnel – Qualification of welding procedures: Validity of a welding procedure qualification*** 18.35

GS Clause 18.35 is changed to GS Clause 18.35(1). The following is added after GS Clause 18.35(1):-

(2) A record of the acceptance test for welding procedure qualification record (WPQR) shall be submitted to the *Supervisor* for acceptance at least 3 weeks before fabrication of the steelwork starts. The record shall be in the form stated in BS EN ISO 15614-1: 2004 + A2 : 2012 and shall be endorsed by the testing consultant accepted by the *Supervisor*.

**MECHANICAL FASTENING**

***Tightening of bolts*** 18.54A

Method statement shall be submitted to indicate the tightening methods, tools, Plant and applied bedding torque to the Supervisor for review without objection.

**GEOMETRICAL TOLERANCES**

***Tolerance types*** 18.67

(9) The tolerance requirements on the vertical alignment are defined below:

*Design vertical alignment:-*

The design alignment is the intended vertical alignment as shown on the drawings.

*Final vertical alignment:-*

The final alignment is the vertical alignment as constructed after application of all permanent loads at reference temperature 20 ℃.

*Construction vertical alignment:-*

The construction alignment is the intended vertical

alignment at the completion of the Works, i.e. before application of certain permanent loads like wearing surface, barriers, lighting etc.

(10) The final alignment shall be of the same basic shape as the design alignment and within the tolerances as shown in the Section 9 of the Specifications.

(11) The *Contractor* shall determine the tolerances for the construction alignment which takes into account the variations in stiffness and weight and which will satisfy the requirements of the final vertical alignment. The *Contractor* shall submit his calculation to the *Supervisor* for agreement at least 6 weeks in advance of any bridge steelwork erection.

**INSPECTION,TESTING AND CORRECTION**

***Constituent products and components – Constituent products***

18.72 GS Clause 18.72(1)(a) is deleted and replaced by:-

(a) A batch of steelwork is the amount of steelwork delivered to the fabricator from a single cast or heat.

GS Clause 18.72(1)(c) is deleted and replaced by:-

(c) Samples of steel shall be provided from each cast/heat of steel within 3 days after delivery to the fabricator's *works* or to the Site. The rate of sampling and the position and direction of the samples shall be in accordance with BS EN 10025 Part 1 – 6: 2004 or BS EN 10210-1 : 2006 as appropriate.

The rate of sampling for the steel material under the scheme of Inspection Report 3.2 shall be carried out on an average every fourth cast/heat of steelwork, as directed by the *Supervisor*.

(6) At the low points of each continuous length of sealed trough stiffener or similar cavity holes, removable plugs shall be provided for drainage. During construction the plugs shall be removed at intervals to check for water. If water is found in any trough, all the plugs in that group shall be removed. In the completed bridge the plugs shall be removed except plugs sealing holes to the exterior.

***Pre-production procedure trials***

18.73A (1) Pre-production procedure trials shall be carried out for the following elements without exceptions:

(a) 10mm or more thick and heavily restrained welds

(b) welds between connection plate

(c) Welds on connection of main members

(d) stainless steel plate

The pre-production procedure trials shall demonstrate that the specified penetrations are achieved.

The restraint conditions in the trials shall resemble the condition in the field.

A WPQR (Welding Procedure Quality Record) shall follow

with each test containing information as required in applicable relevant standards.

***Inspection authority and testing consultant*** 18.81A The testing consultant for non-destructive testing of welds shall be a specialist contractor included in the category of 'Specialized Operations for Highway Structures (Class IV)' in the List of Approved Suppliers of Materials and Specialist Contractors for Public Works, to be appointed by the *Contractor* and accepted by the *Supervisor*.

***Testing: welds*** 18.82A (1) Welds for structural steel and steel castings shall be visually inspected in accordance with BS EN ISO 17637: 2011. After visual inspection, non-destructive testing shall be carried out in accordance with table below. The compliance criteria shall be as stated in Level B of BS EN ISO 5817: 2014.

| **Weld Type** | **Frequency of Non-destructive Testing** |
|---|---|
| All types of butt welds, including full and partial penetration welds | 100% ultrasonic examination and magnetic particle inspection (Welds inspected by ultrasonic examination shall be spot checked by radiographic inspection / in 5% of ultrasonic examination inspected length or as directed by the Supervisor) |
| All types of fillet welds | 40% magnetic particle inspection / dye penetration test |
| Secondary attachment welds, e.g. for fixing purlins, side rails, E&M hangers, etc. | 20% of attachment by magnetic particle inspection / dye penetration test |

Whenever a hard imperfection is disclosed, the cause of the imperfections shall be investigated and corrective actions shall be implemented. The re-examination of the relevant welder qualification is required if imperfections exceeding the limits by ≥3% of the welded weekly length for that concerned welder.

In the case of soft imperfections exceeding the limits by ≥3% of the welded weekly length based, the inspection extent of that weld number shall be doubled in the unit intensity of 100% is reached.

(2) Non-destructive testing of full and partial penetration butt-welds shall be carried out by ultrasonic examination in accordance with BS EN ISO 17640: 2010 by radiographic examination in accordance with BS EN ISO 17636-1: 2013 or BS EN ISO 17636-2: 2013. Non-destructive testing of fillet welds shall be carried out by either penetrant method in accordance with BS EN ISO 3452-1: 2013 or magnetic particle flaw detection method in accordance with BS EN ISO 17638: 2009. Welds shall be dressed to facilitate ultrasonic examinations.

(3) The *Contractor* shall produce NDT-drawings and submit to the *Supervisor* for agreement, in which define each weld with regard to:

(a) weld number

(b) weld type,

(c) weld class,

(d) NDT- inspection type,

(e) NDT- inspection intensity.

The *Contractor* shall prepare an Inspection Plan. The plan shall as a minimum contain the following:

(i) A system for identification of field documentation.

(ii) Plans of registration, which make it possible to trace the field documentation back to the individual parts of the structure as well as to read the immediate inspection status.

(iii) An account of which test procedures/methods/extent are valid for each type of weld.

(iv) A division of the welding work in inspection groups.

(v) Rules for extension of inspection and repeated inspection if non-conforming welds are detected.

(vi) Rules for lowering of inspection.

(vii) Succession of inspection and testing activities.

(viii) Hold points where inspection and testing shall be completed before production may proceed.

(4) Welds for steelwork which has been fabricated and tested by non-destructive testing at the fabricator's works shall be visually inspected for cracks when the steelwork is delivered to the Site. 5% of the welds shall be examined for cracks by the magnetic particle flaw detection method; or other accredited NDT method as approved by the *Supervisor*.

(5) An Inspection Plan shall be submitted to the *Supervisor* for acceptance describing in words and schedules the proposed method for carrying out non-destructive testing on welds. The NDT-method most appropriate for disclosing the type of imperfections most commonly seen shall be used for the inspections.

The proposed Inspection Plan shall follow the general rules of BS 5400-6 and BS EN 25817, supplemented with the further requirements given in these Particular Specifications as follows:

(a) quality requirements for welding shall be in accordance with Level B of BS EN ISO 5817, supplemented with the further requirements of

Table 18.4S.

(b) external discontinuities on the weld, such as sharp melt hollows, incomplete butt welding, visible pores and shrinkage hollows shall be smoothed or removed.

(c) welding switch-on scars, weld spatter and slag remains shall be avoided or removed.

(d) representative test pieces shall define the acceptance quality shall be produced and retained throughout the project for comparison with production welds.

(6) Production tests including tensile tests and charpy Vnotch impact tests shall be conducted on full penetration butt weld joints and T-butt joints in reorientated position specimens at the following sampling tests:

(a) one sample on every 10 nos. of joints at tension zones

(b) one sample on every 20 nos. of joints in compression zones

**SUBMISSIONS**

***Particulars of constituent products supplied for use in the works***

18.86 (2) Inspection Reports shall be directly submitted from the Inspection Authority to the Supervisor not more than 2 days after the steel has been delivered to the place of fabrication.

(3) All stainless steels shall be supplied with a mill certificate recording test results for the chemical composition and mechanical properties as given in BS EN 10088 Parts 1 to 3.

(4) The *Contractor* shall submit to the *Supervisor* for acceptance a Quality Plan for inspection and testing of the steelwork and associated components. The Quality Plan shall describe in words and schedules the intended testing and inspection procedures to be followed for the works. The results of testing and inspection shall be registered in a way, that the following requirements to traceability are met:

**Table for requirements of traceability**

| **Item Specification** | **Traceability** | |
|---|---|---|
| Clause | Limited | Complete |
| Inspection and testing | | x |
| Steel materials | | x |
| Stainless steel materials | | x |
| Cast Steel | | x |
| Bolts, screws, nuts and washers | x | |
| Inspection for lamination | x | |
| Inspection of preparation | x | |

| | | |
|---|---|---|
| Inspection of handling of welding consumables | x | |
| Inspection of the execution of welds | x | |
| Identification of welders | | x |
| Inspection of welds | | x |
| Testing for tightness | x | |
| Surface inspection at removed attachments | x | |
| Damages during transportation and at receiving | x | |
| Inspection of tightening of bolts | x | |
| Deviations in adjoining structures | | x |
| Inspection of geometrical imperfections | x | |
| Inspection of main geometry and installation | | x |

***Particulars of welding procedures*** 18.87

(d) welding procedures in accordance with BS EN 1011 -1 : 2009 and BS EN 1011 -2 : 2001.

(e) Procedure trials shall, without exceptions, be carried out for 10mm or more thick and heavily restrained welds, including welds between main members.joints or welds, which are necessary for the fabrication, delivery and erection of steelwork, but not specified in the contract;

***Particulars of workshop drawings*** 18.90

(1)(a)***S*** Steelwork and welds, including any stud welds, marked with the relevant welding procedures, the sequence of which shall be employed to minimise contraction stresses and distortion. The accepted welding procedures shall also be fully detailed on the shop drawings and strictly adhered to in production.

(3)***S*** Two hardcopy in A3 size and CD-ROM (containing digital data files in .pdf and .dgn format) of the accepted drawings shall be submitted to the Supervisor at least 7 days before fabrication of steelworks start.

(4) Submission of shop drawings for the acceptance by the *Supervisor* shall indicate that the general requirements of the design have been satisfied. Acceptance shall not be taken to imply that the dimensions or the correct matching of connections or joints have been checked and these will remain the responsibility of the *Contractor*. Drawings submitted for acceptance shall be of a good standard agreed by the *Supervisor*. Any drawings which require amendment will be returned for alteration with a detailed review of all the corrections and re-submission.

(5) The *Contractor* shall be responsible for any amendments of the steelworks design to cater for their proposed

method of fabrication, delivery and erection. The amendments with supporting design calculation and appropriate checking certificates shall be submitted to the Supervisor for acceptance at least 3 weeks before fabrication of the steelwork starts.

(6) Fabrication shall not commence until the *Contractor* has received the *Supervisor*'s acceptance of this proposals.

***Particulars of delivery of steelwork*** 18.91 (3) The *Contractor* shall prepare and submit to the Supervisor for acceptance a method statement containing complete plans and procedures for lifting, loading and transportation and erection of the structural steelwork.

***Particulars of method of erecting steelwork*** 18.92 (2)***S*** The Method Statement shall be submitted to the *Supervisor* at least 12 weeks before erection of the steelworks starts.

(3) Temporary attachments required for handing and erection shall be made using the same grade of steel as that used for structure. Where possible these attachments shall be made on a surface that is not visible on the finished structure, where this is not possible the surface finish on the steel shall be reinstated to the specified level of finish after removal of the temporary attachment. Magnetic Particle Inspection (MPI) / Penetration Test (PT) shall be conducted on all the removal area of temporary attachments to ensure there are no cracks in the surface of parent metal.

***Particulars of welders*** 18.95 (1)***S*** Welding certificates endorsed by the approved inspection authority or testing consultant shall be submitted to the *Supervisor* to show that each proposed welder has been accepted in accordance with BS 4570 : 1985, BS EN 287-1 : 2011, BS EN ISO 9606-1:2013 or BS 4872 : Part 1 : 1982 as appropriate. The extent of acceptance of the welder shall be appropriate to the categories of welds the welder will carry out.

***Particulars of inspection authority and testing consultant*** 18.97***S*** (1) The name of the proposed inspection authority issuing the Inspection Report 3.2 in accordance with BS EN 10204 shall be accredited to NF EN ISO/CEI 17020:2012 or International Federation of Inspection Agencies (IFIA) and permitted by the *Supervisor*.

(2) The name of the proposed testing consultant endorsing welder certificates, records approval tests for welding procedures and carrying out and documenting nondestructive inspection of the finished welds shall be accredited to the HOKLAS laboratory as permitted by the *Supervisor*.

(3) Personnel who perform non-destructive testing or assess the results shall meet the requirements for certification as "welding inspector" according to document "Certification Scheme for Welding Inspection Personnel

(CSWIP)" or American Welding Society (AWS).

***Particulars of paint*** 18.98 (1)(d) an undertaking by the *Contractor* and the painting manufacturer that they will issue the warranty in the format as specified in PS Appendix N1.

***Samples of Materials*** 18.99 (4) Paint samples, sufficient to fulfil the requirements as specified in PS Appendix O, shall be provided by the *Contractor* for testing in the laboratory approved by the *Supervisor*.

**FABRICATION OF STEELWORK**

***Fabrication of steelwork*** 18.100 (1) (a) Fabrication of steelwork shall comply with BS EN 1090: Part 2.

(b) Variations to the accepted drawings shall not be made except with the acceptance of the *Supervisor*.

(c) The drawings shall show full and complete information regarding the size, location and type of welds and the sequence which is to be employed to minimise contraction stresses and distortion. The accepted welding procedures shall also be fully detailed on the shop drawings and strictly adhered to in production.

(d) Acceptance by the *Supervisor* of shop drawings shall indicate that the general requirements of the design have been accepted. Acceptance shall not be taken to imply that the dimensions or the correct matching of connections or joints have been checked and these will remain the responsibility of the *Contractor*. Drawings submitted for acceptance shall be of a good standard and any drawings which clearly require considerable amendment will be returned for alteration with a detailed review of all the corrections which may be required.

(2) The *Contractor* shall select a sequence of fabrication which shall keep distortion to a minimum. An allowance shall be made for the shrinkage of the end to end butt welds in the main beams of 1.5mm for each such weld.

(3) Fabrication shall be carried out under controlled conditions which afford protection against dust, dirt and adverse weather, including the cooling effect of wind on welds between members of high yield steel. Welding shall not be carried out when the workpiece is wet or when the temperature falls below 2°C.

(4) Prior to the acceptance of fabricated steelwork sections trial assembly shall verify correct geometry once erected. Prior to commencement of the trial assembly the *Contractor* shall submit a plan describing the methods and procedures for measuring, recording, and controlling

the geometry of the bridge deck and upper arch trial assemblies. The procedure shall include methods for combining the survey results of each individual trial assembly to calculate and check the cumulative geometry of the complete structure assembly as trial assembly proceeds. The support conditions in the trial assembly shall resemble the support conditions in the field. The sections shall have identical temperature and be in equilibrium. Correlation to the reference temperature of 22℃ shall be made if the temperature during trial assembly deviates from the reference temperature. Procedures for trial assembly are subject to acceptance by the *Supervisor*. A report shall be kept to monitor deviations in geometry on single sections and accumulated deviations.

(5) The *Contractor* shall submit details of the site welding proposals, including staging, platforms and weather protection for welders and materials, to the *Supervisor* at least 14 days prior to the commencement of the site welding.

(6) Tack welds to BS EN 1101 shall be not less than 50mm long. Tack welding shall be subjected to prior acceptance by the *Supervisor*.

(7) Welding shall be carried out only by qualified welders and under the direction of an experienced and competent welding QA/QC Supervisor. Unless otherwise agreed by the *Supervisor*, a record shall be kept to enable major butt welds to be identified with the welders responsible for the work. Finished works shall not be hard stamped for this purpose.

***Reinstatement on zinc coating with post-galvanising welding***

18.101 (1) Where post-galvanising welding is necessary, the zinc coating shall be ground off the mating surfaces directly prior to welding. Immediately after welding, the surfaces of the weld area shall be prepared by removal of slag with the chipping hammer followed by vigorous wire brushing. The zinc coating shall then be restored by either (a) or (b) below:

(a) The welded joints shall be zinc sprayed in accordance with BS EN ISO 2063 : 2005 to a minimum thickness of 0.2mm. Two coats of an organic zinc rich paint complying with BS 4652 : 1995 shall then be applied across the entire zinc sprayed areas including at least 25mm of the parent galvanised coating.

(b) Pre-heating to 315 °C and application of proprietary metallic repair stick or powder to a thickness greater than 0.2mm.

(2) If re-coating does not take place within 4 hours of welding, the *Supervisor* may require the weld area to be vacuum grit blasted to BS EN ISO 8501 ‘first quality’ and

hot-zinc spray coated. Work shall be in accordance with BS EN ISO 2063 : 2005 and to a minimum coating thickness of 0.2mm.

(3) After reinstatement of zinc coating, a complete paint system shall be applied to the repaired area in such a manner that the new paint overlaps the existing coats by at least 50mm around the affected part.

***Loading fabricated steelwork***

18.102 The *Contractor* shall not at any time load any structure or any part thereof in excess of the designed working load. No loads shall be placed on any parts of the structure without the prior acceptance of the *Supervisor*

**TOLERANCES**

***Tolerances: fabrication of steelwork***

18.103 The materials for the works and the fabrication, assembly and erection of the works shall be to the tolerances stated in the Specification. Fabrication tolerances for steelwork shall comply with BS 5400: Part 6, Clause 4.2.

However, fabrication tolerances in respect of camber for steelworks complying with BS 5950: Part 2: 2001, subsection 7.2 shall be amended as follows.

The deviation from specified camber ordinate at the mid-length of the portion to be curved, shall not exceed:

(a) 0,

(b) 12 mm or 4 mm/m length of the test length, whichever is greater.

[BLANK PAGE]